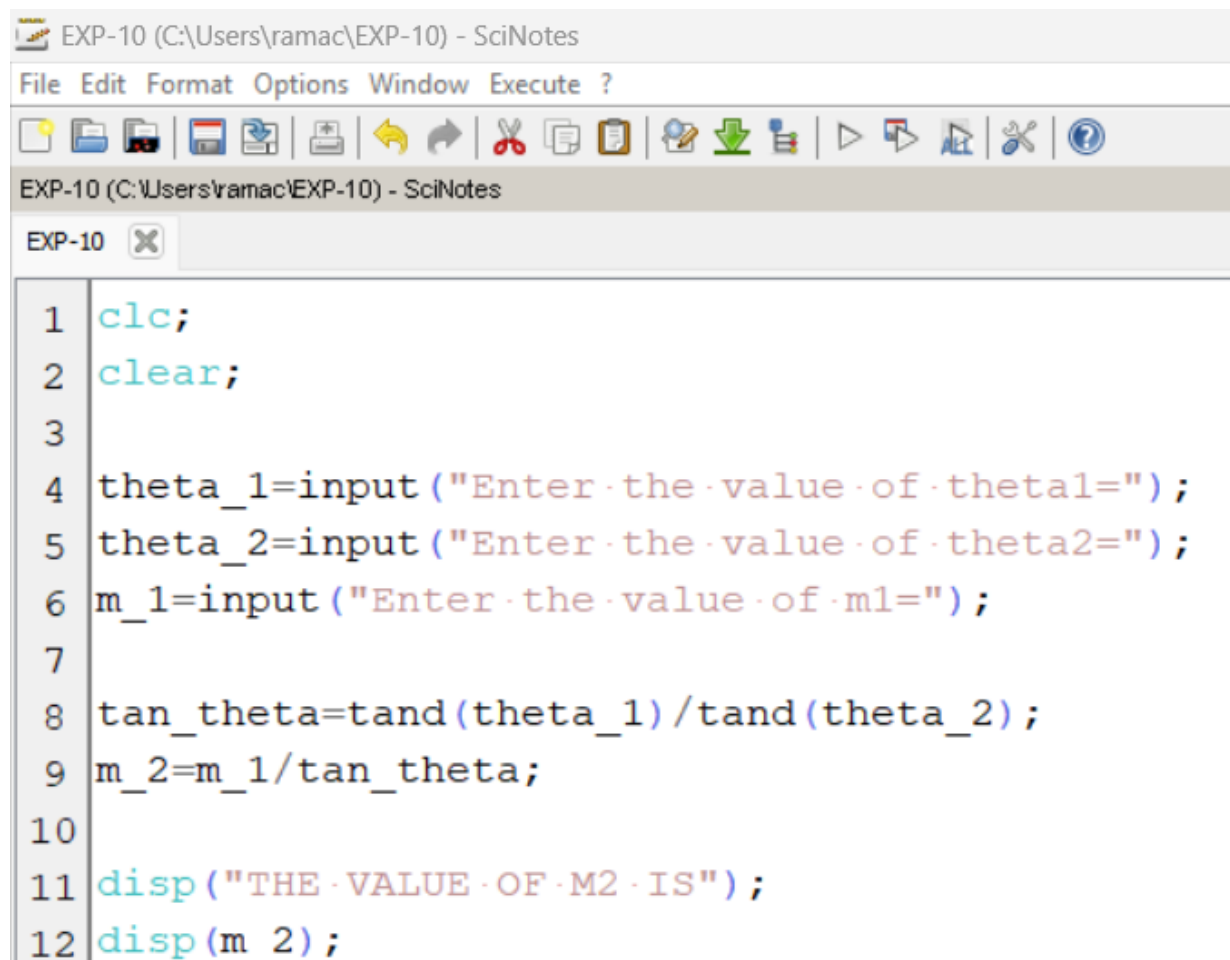


Exp:10 Implement Boundary condition between two different dielectric medium in Magnetic Field and find the relative permeability of the different medium.

Aim: To write program for Boundary condition between two different dielectric media in magnetic field using SCILAB.

Software required: SCILAB version 6.0.1

Practical output:



```
EXP-10 (C:\Users\ramac\EXP-10) - SciNotes
File Edit Format Options Window Execute ?
EXP-10 (C:\Users\ramac\EXP-10) - SciNotes
EXP-10 [X]

1  clc;
2  clear;
3
4  theta_1=input("Enter the value of theta1=");
5  theta_2=input("Enter the value of theta2=");
6  m_1=input("Enter the value of m1=");
7
8  tan_theta=tand(theta_1)/tand(theta_2);
9  m_2=m_1/tan_theta;
10
11 disp("THE VALUE OF M2 IS");
12 disp(m_2);
```

```
Scilab 6.1.1 Console

File Edit Control Applications ?

Scilab 6.1.1 Console

Enter the value of theta1=30

Enter the value of theta2=40

Enter the value of m1=5

"THE VALUE OF M2 IS"

7.2668160

-->
```

Theoretical Output:

⑩ $\theta_1 = 30^\circ$
 $\theta_2 = 40^\circ$
 $m_1 = 5$

$$\tan \theta = \frac{\tan \theta_1}{\tan \theta_2}$$
$$\Rightarrow \frac{\tan 30^\circ}{\tan 40^\circ}$$
$$\Rightarrow \frac{0.57735}{0.83910}$$
$$\Rightarrow 0.6879$$
$$m_2 = \frac{m_1}{\tan \theta} = \frac{5}{0.6879} = 7.274$$

≈ 7.27

Result :

Thus, the value of M_2 was calculated for $\theta_1 = 30^\circ$, $\theta_2 = 40^\circ$, and $M_1 = 5$.
The value of M_2 is found to be 7.27.